

SMART INDIA HACKATHON 2026



TADIT (तड़ित)

AI nowcasting of thunderstorms & lightning, 0–6 hours ahead

- **Problem Statement ID** – [SIH26072](#)
- **Problem Statement Title**- [AIML Based Nowcasting of Thunderstorm and Lightning](#)
- **Theme**- [Disaster Management](#)
- **PS Category**- [Software](#)
- **Team ID**-
- **Team Name** (Registered on portal)



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TADIT: AI LIGHTNING NOWCASTING



Tadit (तड़ित = lightning) fuses radar, satellite, lightning and weather-model data into one AI that forecasts where lightning will strike, every 10 min, up to 6 h ahead.

[Live prototype: tadit.pages.dev](https://tadit.pages.dev)

Proposed Solution



Multimodal AI fusion: one model reads Doppler radar, INSAT / GK2A satellite imagery, lightning strikes and GFS weather-model fields together.



Seamless 0–6 h: Tier 1 deep-learning nowcast every 10 min for 0–3 h, then Tier 2 AI-corrected NWP for hours 4–6, switching at the measured crossover hour.



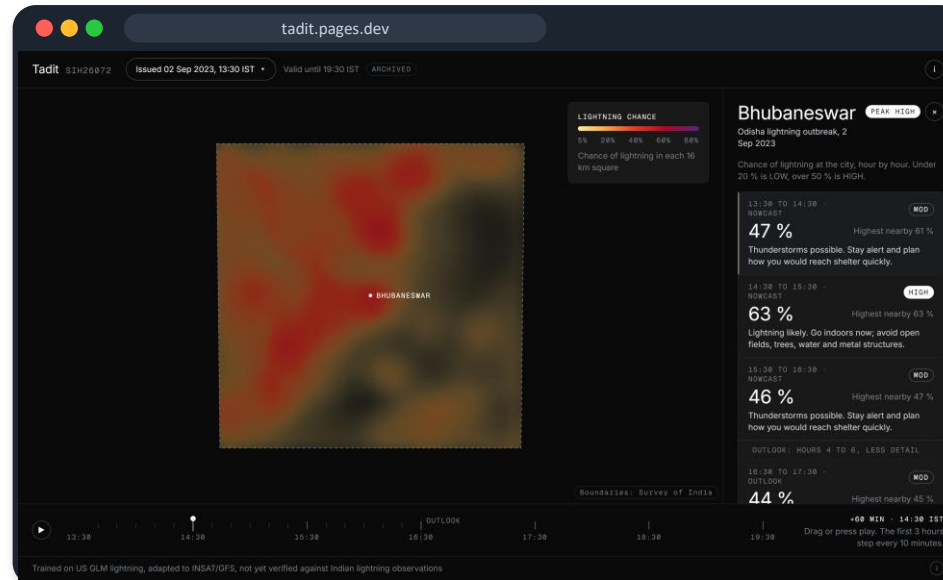
What it delivers: lightning probability per 16 km cell, radar-like storm intensity at 2 km, and LOW / MODERATE / HIGH risk with plain advice for each city.



Built for India's gaps: keeps working without radar or a lightning feed; runs today on INSAT-3DR/3DS or GK2A satellite + GFS.



Working prototype: the Tadit web app shows an animated India map and hourly city risk, live at tadit.pages.dev.



Working prototype: Bhubaneswar lightning outbreak, 2 Sep 2023, forecast issued 13:30 IST (satellite + GFS, no radar)



Observe
radar, satellite,
lightning, NWP



Fuse
one multimodal AI
model



Forecast
0–6 h, every 10 min



Alert
city risk + advice

0.613

lightning CSI at hour 1, 1.8x
persistence

6 / 6

lead hours beating raw NWP
lightning

India

beats persistence vs FY-4A satellite
lightning

How It Addresses the Problem



3 h → 6 h

IMD's nowcasts look 3 h ahead. Tadit forecasts **6 h**, refreshed every 10 min.



No radar? Still works

Large parts of India lack radar. Satellite + NWP alone reaches **0.461** CSI at hour 1.



From maps to action

City-level hourly risk with advice (“go indoors now”) for officials and the public.



Innovation & Uniqueness

1

Mid-fusion U-Net with NWP cross-attention + FiLM: all 18 lead times in one pass.

2

Modality dropout: robust to any missing sensor, with free drop-a-sensor tests.

3

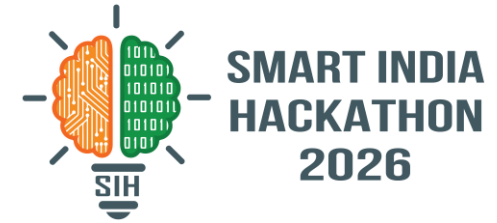
Measured tier crossover stitches nowcast and NWP into one 0–6 h product.

4

India-adapted in training (INSAT / GFS-like inputs) with leak-free lightning labels.

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TECHNICAL APPROACH



System architecture: from raw observations to a 0–6 h lightning forecast

1 · DATA SOURCES



Radar

IMD DWR · NEXRAD VIL



Satellite

INSAT-3DR/3DS · GK2A · GOES IR



Lightning

ILDN / strike feeds · GLM



Weather model

GFS 0.25° · HRRR

2 · PREPROCESSING



384 km tiles on one map grid



Radar / satellite at 2 km, lightning at 8 km



Leak-free lightning labels (past 5 min only)



NWP only from runs issued before the forecast



50% of samples degraded to INSAT & GFS quality

3 · AI MODELS

TIER 1 · Fusion nowcaster (0–3 h, every 10 min)

60M params · modality dropout

Per-source encoders

each sensor at native res.

Gated fusion

+ 'missing' embeddings

U-Net + attention

NWP cross-attn · FiLM

18 lead times

VIL 2 km · lightning 8 km

TIER 2 · AI-corrected NWP (lead hours 1–6)

GFS / HRRR forecast fields

Compact U-Net

Lightning + VIL, 16 km

4 · SEAMLESS OUTPUT

Switch at hour 3

Tier 1 for hours 1–3, Tier 2 for 4–6: the crossover measured on held-out storms.



Probability maps



JSON: hourly risk + sources used



Tadit web app & alerts

Methodology



Data: 3,000 real US storm events (SEVIR) + HRRR; split by date, so test storms are unseen.



Train: Intensity-weighted radar loss + focal lightning loss; modality dropout; one night on an RTX 4060 Ti.



Evaluate: CSI / POD / FAR per lead hour vs persistence and raw NWP; drop-a-sensor tests.



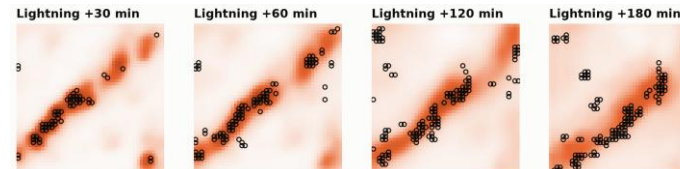
India: Run on INSAT / GK2A + GFS; checked against FY-4A satellite lightning.



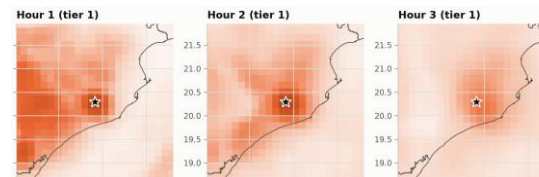
Serve: Maps + JSON exported to a static web app on Cloudflare Pages.

Forecasts vs Reality

Held-out US storm: forecast lightning chance (shading) vs observed flashes (o)



India, Bhubaneswar 2 Sep 2023: hours 1–3 from satellite + GFS only (★ city)



Technology Stack

AI / ML

Python

PyTorch

NumPy

bf16 + torch.compile

DATA

xarray

h5py

s3fs

Herbie

pyproj

cfgrib

WEB

React

TypeScript

Vite

MapLibre GL

INFRA

AWS Open Data

Cloudflare Pages

Hugging Face

RTX 4060 Ti / 5090

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FEASIBILITY AND VIABILITY



Technical Feasibility

Built, trained and tested today



Working two-tier AI: 60M-parameter fusion U-Net for 0–3 h plus an NWP post-processor for 3–6 h, trained on 3,000 real storm events.



Fast: a full 0–6 h forecast for an Indian city in **about 1 minute**, data download included.



Light: trained overnight on **one 16 GB consumer GPU**; the deployed model file is 120 MB.



Reliable build: open-source Python / PyTorch stack, covered by **38 automated tests**.



Operational Feasibility

Runs on India's own data



Indian data readers ready: INSAT-3DR/3DS (MOSDAC), IMD Doppler radar, GFS and lightning strike feeds.



Verified on real files: our INSAT reader matches the GK2A satellite on the same scene ($r = 0.87$).



Works with gaps: runs on whichever sources are live, and every output lists what it used.



Easy to plug in: standard map + JSON outputs for IMD's nowcast portal and alert apps; the web viewer needs no backend.



Economic Viability

Low cost, scalable, policy-aligned



No new hardware: reuses radar, satellite and NWP data IMD already collects; open GK2A / GFS as backup.



Near-zero serving cost: forecasts are static files served from a free CDN (Cloudflare Pages).



Scales simply: ≈ 30 tiles of 384 km cover India; at ~ 1 min per tile, one GPU refreshes the whole country in about 30 min.



Policy fit: supports MoES's **Mission Mausam** (₹2,000 cr) goals of more frequent nowcasts and no undetected weather events.



Potential Challenges & Risks

01

Limited access to data: Indian lightning archives (ILLN / ILDN) and radar volumes are not openly available, so early findings may be misleading.

02

Trained on US storms, used in India: monsoon storms, terrain and INSAT's slower scans differ from the US training data.

03

Sparse radar and sensor outages: radar does not cover all of India, and feeds drop out during storms.

04

Calibration and false alarms: probabilities are tuned to US storm frequency, not Indian climate.

05

Adoption into IMD operations: a new tool must fit forecasters' workflows and existing alert channels.



Strategies to Overcome Them



Gets better as validated data arrives: ILDN data is requested and every output carries an honest label. **As the model gets validated Indian data, it recalibrates and becomes more accurate.**

01



India-mode training: half of all samples are degraded to INSAT (4 km, 15–30 min scans) and GFS (~ 24 km) quality; fine-tune on Indian cases as they come in.

02



Modality dropout: the model runs on whatever is live. Without radar, hour-1 CSI only dips from **0.613** to **0.592**; satellite + NWP alone still scores **0.461**.

03



Regional recalibration with IMD, and alerts issued as risk levels, not raw numbers. On a dry Delhi day the forecast stayed at **0.04**.

04



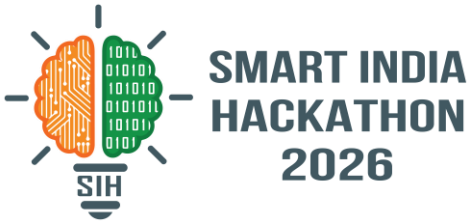
Phased rollout: pilot at one regional centre (e.g. Bhubaneswar) side by side with current nowcasts, then scale state by state.

05

CSI = critical success index for lightning (higher is better), on held-out 2019 US storms. Trained on US GLM lightning, adapted to INSAT/GFS, not yet verified against Indian lightning observations. Tile count is an estimate from India's land area.

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IMPACT AND BENEFITS

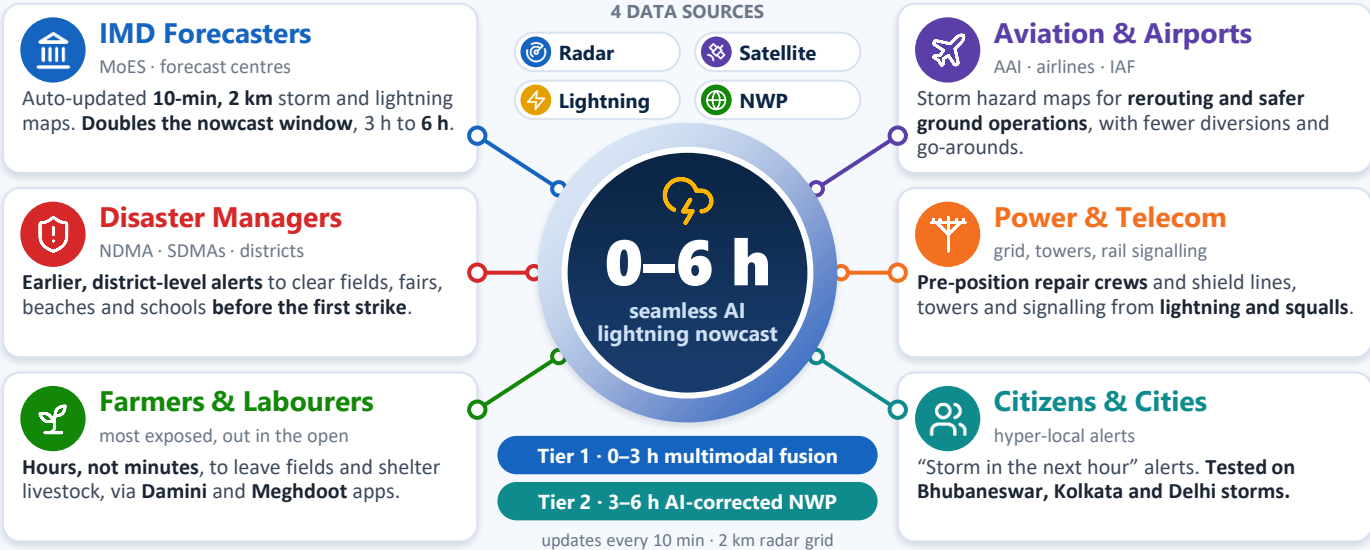


⚡ **2,558 lives**

lost to lightning in India in 2023: 39.7% of all deaths from forces of nature, the largest single share (NCRB). Today's operational nowcasts look only 3 hours ahead.

PS SIH26072 · MoES / IMD
Theme: Disaster Management

Potential Impact on Target Audience



Benefits of the Solution

- Social**
Saves lives where lightning strikes hardest: **MP, Bihar, Odisha, UP and Jharkhand** account for **59%** of India's lightning deaths. Timely alerts build **public trust** in IMD warnings.
- Economic**
Cuts losses to **crops, livestock, flights and power grids**. Built on **existing and open data** (IMD radar, INSAT, GFS): **no new sensors** are needed, and the model runs on a **single GPU**.
- Environmental**
Flags **lightning-prone zones** for **forest-fire watch**, supports **climate adaptation** as severe convective weather grows, and helps **solar and grid operators** plan for storms.
- Strategic**
Atmanirbhar: runs on India's own **INSAT-3DR/3DS** satellites and GFS, and **keeps forecasting when radar or lightning feeds drop out**, so it covers districts **beyond radar range**.

Measured, not claimed

Held-out 2019 storms scored against real satellite lightning (GLM).

1.8×

better than persistence at hour 1

Lightning CSI 0.613 vs 0.341; the gap grows to 5.5× by hour 3

+70%

over raw NWP lightning at hour 6

CSI 0.297 vs 0.175, and ahead of NWP at every hour from 1 to 6

0.461

hour-1 CSI with no radar

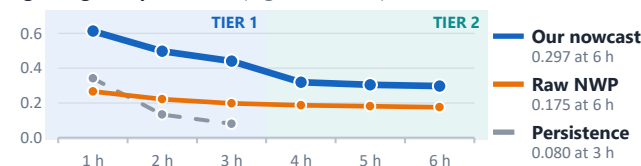
Satellite + NWP only: the usual case outside Indian radar coverage

0.71 vs 0.04

India: storm vs dry day

Storm probability 1 h ahead: Bhubaneswar outbreak (2 Sep 2023) vs Delhi dry day

Lightning CSI by lead hour (higher is better)



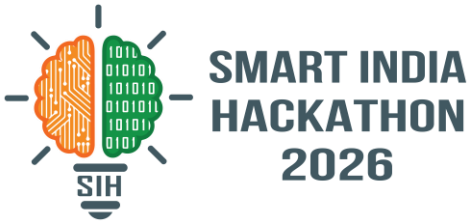
“From a 3-hour to a 6-hour lightning warning: one seamless AI nowcast, built on India's own satellites.”



CSI = critical success index for lightning in a 16 km cell during each lead hour; Tier 1 is used for hours 1–3 and Tier 2 for hours 4–6. India probabilities from GK2A satellite + GFS input, no radar. Trained on US GLM lightning, adapted to INSAT/GFS, not yet verified against Indian lightning observations. Death figures: NCRB, Accidental Deaths & Suicides in India 2023.

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RESEARCH AND REFERENCES



Data Sources

TRAIN

SEVIR storm events: radar VIL, GOES-16 IR, GLM lightning. Veillette et al., NeurIPS 2020 · [registry.opendata.aws/sevir](#)

TRAIN

HRRR US weather model (NWP input + baseline), NOAA · [registry.opendata.aws/noaa-hrrr-pds](#)

INDIA

GFS 0.25° global weather model, NOAA · [registry.opendata.aws/noaa-gfs-bdp-pds](#)

INDIA

INSAT-3DR Imager L1B (3RIMG_L1B_STD), ISRO / SAC · [mosdac.gov.in](#)

INDIA

GK2A AMI L1B satellite, KMA via NOAA, no login · [registry.opendata.aws/noaa-gk2a-pds](#)

VERIFY

FY-4A LMI corrected lightning events, ESSD 18, 5773 (2026), CC-BY 4.0 · [paper](#) · [data](#)

TESTED

ISRO Cherrapunji DWR (RSCHR_L2B_STD), tested, not used · [mosdac.gov.in](#)

MAPS

Survey of India boundaries (OVSF/1M/7) · [onlinemaps.surveyofindia.gov.in](#)

ASKED

ILDN Indian Lightning Detection Network, CeLTS, Tripura University: data requested · [ildn.in](#)

Methods & Research Papers

MODEL DESIGN

- Ronneberger et al., **U-Net**, MICCAI 2015 · [arXiv:1505.04597](#)
- Perez et al., **FILM** conditioning, AAAI 2018 · [arXiv:1709.07871](#)
- Leinonen et al., **Thunderstorm nowcasting with deep learning: multi-hazard data fusion**, GRL 2023 (closest prior work) · [doi](#)

DEEP-LEARNING NOWCASTING

- Shi et al., **ConvLSTM** precipitation nowcasting, NeurIPS 2015 · [arXiv:1506.04214](#)
- Ravuri et al., **Skilful nowcasting with deep generative models**, Nature 597 (2021) · [doi](#)
- Gao et al., **Earthformer** (SEVIR benchmark), NeurIPS 2022 · [arXiv:2207.05833](#)

SENSORS & WEATHER MODELS

- Dowell et al., **HRRR Part I**, Weather & Forecasting 37 (2022) · [doi](#)
- Goodman et al., **GOES-R GLM**, Atmos. Research 125 (2013) · [doi](#)
- NOAA/NASA, **GLM Lightning Cluster-Filter ATBD** (330 ms / 16.5 km flashes) · [pdf](#)
- Blaylock, **Herbie** NWP download tool, Zenodo · [doi](#)

OUR WORK

- Tadit** frozen results and scorecard, [results/RESULTS.md](#) · live site [tadit.pages.dev](#)

VERIFICATION & PHYSICS

- Schaefer, **Critical success index** as a warning-skill score, Weather & Forecasting 1990
- Greene & Clark, **Vertically integrated liquid (VIL)** from radar, Monthly Weather Review 1972
- Maddox, **Mesoscale convective complexes** (−52 °C cloud-top threshold), BAMS 1980



Live prototype: click here
[tadit.pages.dev](#)



Code & results: click here
[github.com/adit1290codesrank/sih2026](#)



India Check vs Real Lightning

FY-4A satellite lightning, CSI for hours 1 / 2 / 3 (persistence in brackets)

Bhubaneswar · GK2A **0.24 / 0.22 / 0.20** (0.20 / 0.01 / 0.00)

Bhubaneswar · INSAT **0.24 / 0.16 / 0.13** (0.20 / 0.01 / 0.00)

Guwahati · INSAT **hour 6: MODERATE risk** storm hit the city then



Real Storms We Forecast

Bhubaneswar, 2 Sep 2023: 61,000 lightning strikes in two hours ([Organiser](#)); 10 killed in six districts ([Deccan Herald](#))
Kolkata, 9 May 2024: thundersquall and hailstorms lash the city ([ThePrint](#))



Government Sources

NCRB, **Accidental Deaths & Suicides in India 2023** (lightning deaths) · [data.gov.in](#)
IMD **station-wise nowcast** service (3 h horizon) · [mausam.imd.gov.in](#)
Cabinet approves **Mission Mausam, 2024** · [pmindia.gov.in](#)